



Identifying Training Gaps Between CAMPEP Residencies and the ABR Part 3 Oral Exam for Medical Physics

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Disclosure

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Visiting Scholars Program

Outline

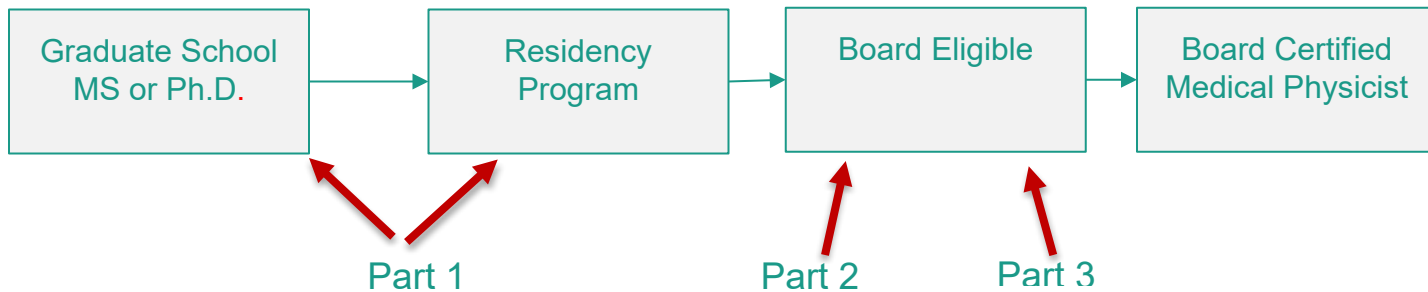


- Project aims
- Overview of the identified research methodology
- Research questions
- Survey
- Result
- Next Steps

Medical Physics Journey

ABR

AMERICAN BOARD
OF RADIOLOGY



Part 3 (Oral) Exam Results

First-Time Takers

Year	Percent Passed	Percent Failed	Percent Conditioned	Total Examinees
2018	60	20	20	228
2019	70	22	8	227
2021	63	23	14	364
2022	79	10	11	121
2023	58	26	16	188
2024	67	14	18	146
2025	63	19	18	145



Brief Overview and Aims of Project

- Identify educational gaps
- Develop a centralized virtual resource database
- Evaluate the feasibility
- Improve accessibility
- Establish a scalable framework

Research Questions



What are the key gaps in preparation for the ABR Part 3 Medical Physics Therapy Track oral exam?

What is the impact of a centralized database of virtual education resources on exam preparedness and performance of medical physicists taking the ABR Part 3 exam?

In what ways can a standardized virtual education platform for the ABR Part 3 exam serve as a scalable model for improving the entire board exam preparation and the maintenance of certification for medical physicists?

Overview of the Identified Research Methodology



- ***Participant Recruitment & Evaluation:***
 - Target: ABR Part 3 Therapy Track examinees (2021–2025)
 - Conduct structured surveys and interviews to identify educational gaps
 - Introduce database prototype and assess usability

Quantitative Methods (Survey Design)



- It contains 62 questions
 - All the answer is multiple choices
- The oral exam has five main topics, and each topic also has subtopics:
 - Reference and relative dosimetry
 - Treatment machines
 - External beam treatment planning, uncertainty management, and treatment planning system QA
 - Brachytherapy, radiation protection, radiation biology
 - Patient safety, data transfer and integrity, professionalism, and ethics
 - a multiple-choice question with five options
 - **Very Adequate, Adequate, Somewhat Inadequate, Very Inadequate**

Qualitative Methods (Interviews)



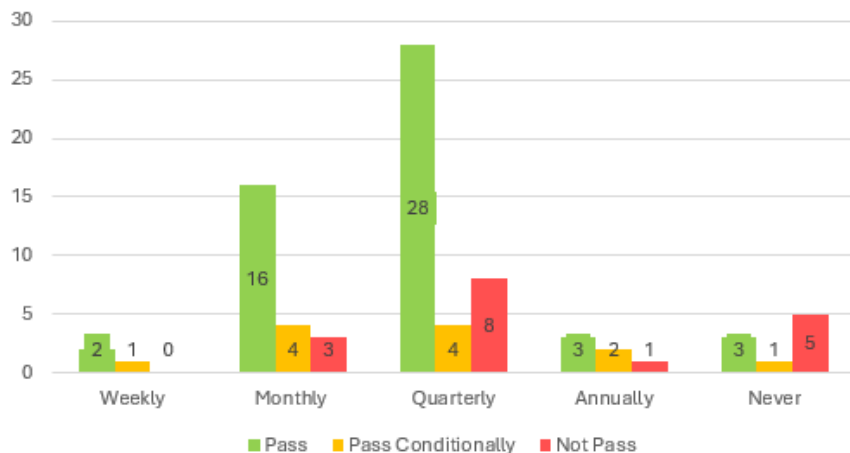
- **Gap during the residency**
 - Mock Exam during residency
 - Mentorship during residency
 - Physics
 - Reaction not cover:
 - Structure
- **Study for the exam**
 - Study Group
 - Web site
 - Salif study
 - Tutoring
- **Suggestion**
 - Videos
 - Pictures
 - Reading
 - Questions Bank
 - comfortable
 - ABR guideline

Quantitative Findings (Survey)

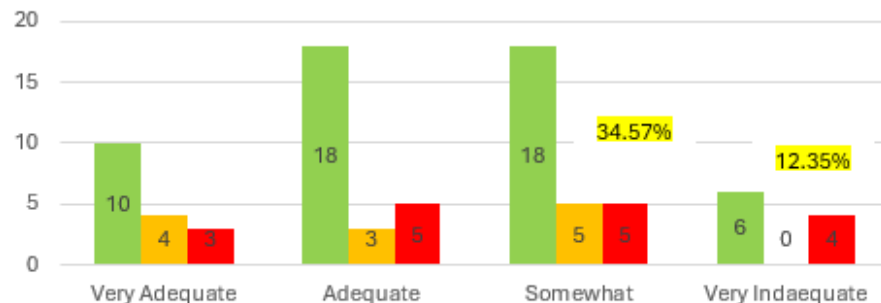
Did you pass the ABR Part 3 Exam on the first attempt?



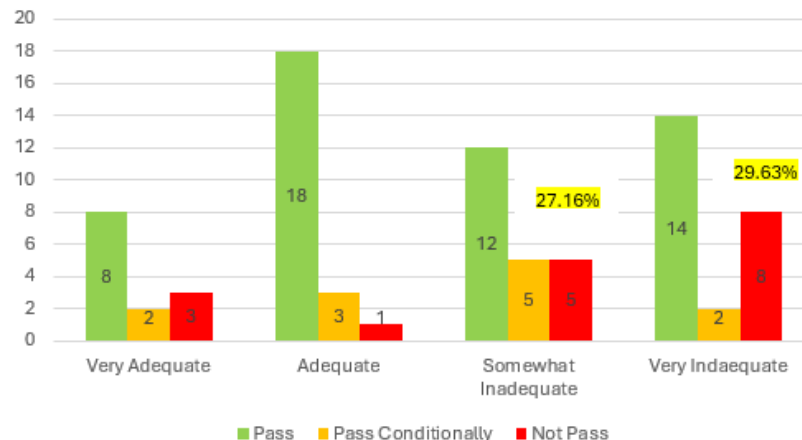
Approximately how often did your program provide mock oral exam-style training?



Specialised delivery units (GammaKnife, CyberKnife, TomoTherapy, MR-Linac).



Proton therapy units (beam delivery, range uncertainty, QA).



Qualitative Findings (Interviews)

Gap during the residency - Mock Exam during Residency

- **Theme 1: Difficulty Calibration Gap**

- FF → "Rotation exams were easy" ,even though she passed
- JJ → "Part 3 questions were in general harder than residency" , direct comparison
- AA → "Not exactly ABR style"
- KK → No time for prep beyond rotation requirements

- **Theme 2: The Availability Problem**

- BB → "Who's available and it's hard to get"
- EE → "Low number of physicists... many not available"
- NN → "Whoever was available that day participated"
- KK → Time pressure eliminated prep entirely

- **Theme 3: High Pressure**

- standing before 5-6 people throwing random follow-up questions. This is actually more stressful than Part 3, yet still "not comprehensive."

Qualitative Findings (Interviews)



Gap during the residency - Mock Exam during Residency

- Theme 4: Structural Barriers: Faculty & Time Constraints
- The quality of preparation depends on who is available, not on a standardized system
 - Supporting responses: BB, EE, KK, NN, EE
- Key patterns:
 - Faculty availability drives exam quality, not program design
 - Low physicist numbers = fewer questioners per session
 - Understaffing and patient load limit mentor engagement
 - *"Whoever was available that day participated"*, not structured

Qualitative Findings (Interviews)

Gap during the residency - mentors

- Theme 1: Variability in mentorship
- *Some mentors provided mock exams*
- *Some explained concepts deeply*
- *Others had limited time due to understaffing*

- Theme 1: Structured
- *II → “Weekly reading discussion and Socratic questioning*
- *EE → mentor dependent*
- *NM: Mentors are far removed from their own exam experience.*

- Theme 1: limited time because we were understaffed
- *EE → “we were understaffed and there were a lot of patients.”*
- *HH → “it is more of the time crunch because we were understaffed.”*
- *AA, II NN positive with mentor, and EE and HH not.*

Qualitative Findings (Interviews)



Gap during the residency
- Inconsistent or Unclear Structure in Residency Training

- CC: explicitly said the residency “wasn’t structured practically.”
- GG: was a resident during COVID, which disrupted normal structure.
- KK: described long clinical hours plus research expectations.
- DD felt the residency “prepared me enough.”

Qualitative Findings (Interviews)



Study for the exam
Study Group

- Theme 1: Collaborative question practice
 - AA: “We prepared questions beforehand... thinking like an examiner.”
 - DD: “Mock exam-like... present questions to my partner.”
 - KK: “You need to be asking questions to each other.”
 - LL: “Mock exams with physicists and peers.”
- Theme 2: Peer to peer knowledge sharing
 - HH: “My partner was able to solve LDR questions... I helped with CyberKnife.”
 - II: Worked with co-resident and a grad school friend.

Qualitative Findings (Interviews)



Study for the exam
Self-Directed Study

- Theme 1: Self recording and self evaluation
 - BB: Recorded themselves answering questions to hear how they sounded.
 - NM: Created flashcard prompts and evaluated their own performance.
- Theme 2: Using AI for mock orals
 - CC: Used ChatGPT/Gemini to simulate oral exam questions.
- Theme 3: two-phase learning strategy
 - FF studied broadly first, then went deep into procedures

Qualitative Findings (Interviews)



Study for the exam
Regional AAPM Chapter Mock Exams

- Theme 1: High value mock exam experience
- **Participants:** AA, BB, CC, EE, JJ, KK.
 - Very similar to the real ABR exam
 - Run by physicists with exam experience
 - Full-length (25 questions or 5 sections)
 - Structured with breaks and multiple examiners
- Theme 2: *easier* than the real Part 3 exam
- **FF:** mock exams did NOT fully match the difficulty of the real Part 3

Qualitative Findings (Interviews)



Study for the exam
Tutoring

- Theme 1: The value of professional tutoring for ABR exam preparation
- **Participants: AA, CC, GG, HH**
 - improved their performance, confidence, and understanding.
 - 5–6 mock sessions
 - Recorded practice
 - Coaching on how to answer concisely
 - Getting used to the exam format
 - Sharing resources

Qualitative Findings (Interviews)

Suggestion: Videos

- Theme 1: Need for high quality video demonstrations
- **Participants:** AA, DD, HH, II, MN
 - Videos showing how machines work
 - Videos showing daily QA
 - Videos showing CyberKnife, Gamma Knife, MR-Linac, protons, etc.
 - Visual identification of machine parts
 - Walk-throughs of special techniques

» Theme 2: mock exam video examples

» **Participants:** LL, JJ

Qualitative Findings (Interviews)



Suggestion: Pictures

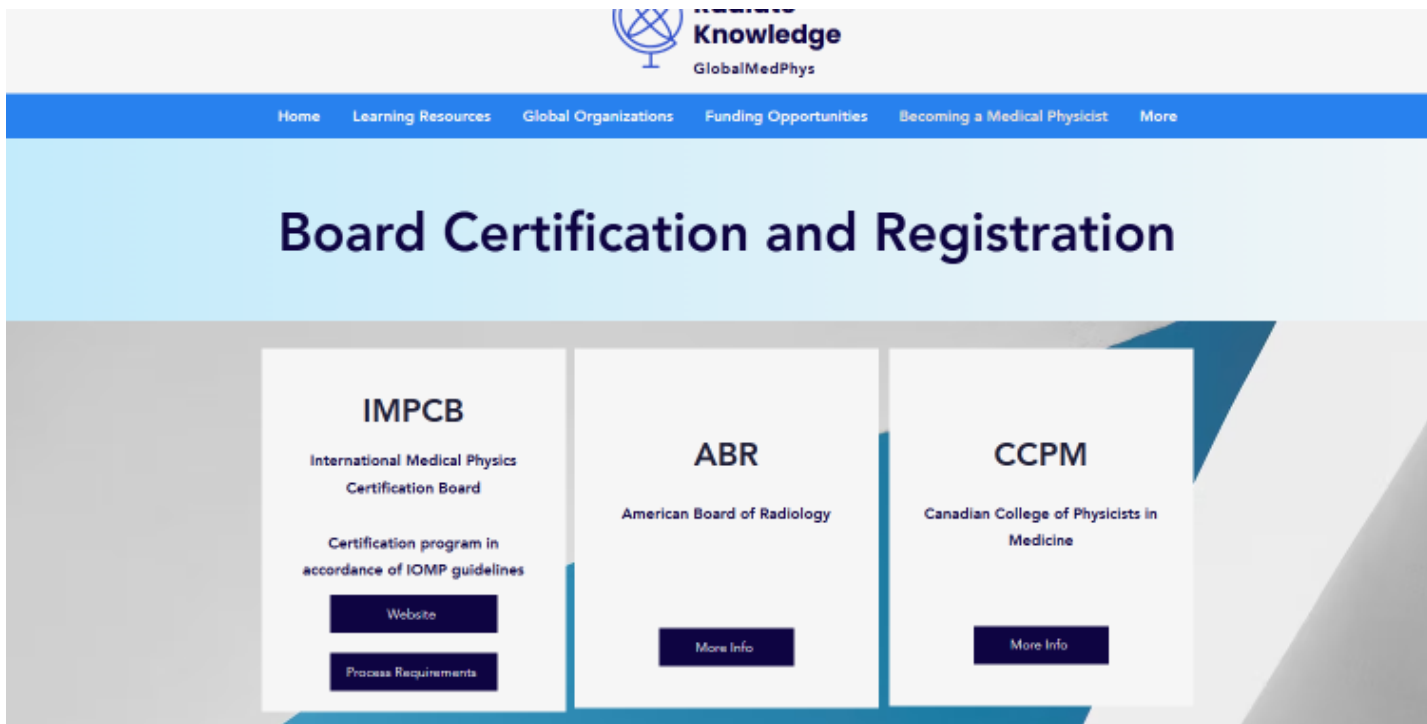
- Theme 1: Visual identification of equipment
- Participants: BB, HH
 - Pictures of applicators
 - Pictures of therapy tools
 - Pictures of HDR equipment
 - Pictures of CyberKnife, Gamma Knife, etc.

Qualitative Findings (Interviews)

Suggestion Reading-Based Learning

- Theme 1: Need for concise summaries
- **Participants:** FF, II
 - “Wikipedia-style summaries”
 - Condensed explanations of long TG reports
 - Key points highlighted
 - Simplified breakdowns of heavy documents
- » Theme 1: Use of question banks for practice
- » **Participants:** HH, KK, LL
 - » Question banks help them get into the exam format
 - » Seeing questions phrased differently improves communication
 - » Practice questions help identify weak areas
 - » Question banks are especially needed for unfamiliar modalities

Next Step



The screenshot shows the top portion of the GlobalMedPhys website. At the top, there is a logo with a globe icon and the text "GlobalMedPhys Knowledge". Below this is a blue navigation bar with links: "Home", "Learning Resources", "Global Organizations", "Funding Opportunities", "Becoming a Medical Physicist", and "More". The main content area has a light blue header with the title "Board Certification and Registration". Below this, there are three white boxes, each representing a different certification board. The first box is for the IMPCB (International Medical Physics Certification Board), which includes a description of the certification program and two buttons: "Website" and "Process Requirements". The second box is for the ABR (American Board of Radiology) and includes a "More Info" button. The third box is for the CCPM (Canadian College of Physicists in Medicine) and includes a "More Info" button.

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[Home](#) [Learning Resources](#) [Global Organizations](#) [Funding Opportunities](#) [Becoming a Medical Physicist](#) [More](#)

Board Certification and Registration

IMPCB

International Medical Physics
Certification Board

Certification program in
accordance of IOMP guidelines

[Website](#)

[Process Requirements](#)

ABR

American Board of Radiology

[More Info](#)

CCPM

Canadian College of Physicists in
Medicine

[More Info](#)

Next Step



AAPM SPEAKER LIST

20+ speakers available for proton therapy



Topics and example speakers:

General

Jess Scholey
Paige Taylor
Austin Faught

Commissioning

Charles Shang
Christopher Ainsley

Treatment Planning

Samantha Hedrick

QA

Chengyu Shi

Shielding

Nisy Elizabeth Ipe

RadBio

Gabriel Sawakuchi

Imaging

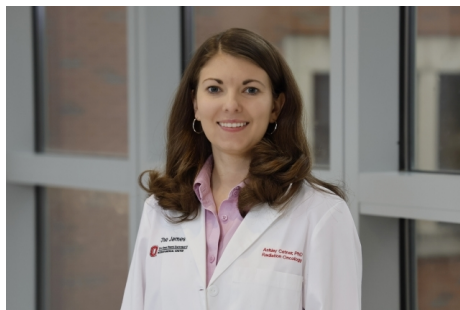
Reinhard Schulte
Jon Kruse

Adaptive

Arthur Lalonde
Brian Winey



Our Team



Thank
You!